

GROUP WORK PROJECT # _1_
GROUP NUMBER: ____7797____

MScFE 632: Machine Learning in Finance

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Step 1

Category 3:

Principal Component Analysis (PCA):

PCA is a statistical or machine learning method for **feature extraction** and **dimensionality reduction**. PCA is a unsupervised method as we don't make use of the target labels or variables.

A transformation is applied to the original data such that a new coordinate system is generated where the first few coordinates, called the principal components, can explain the maximum possible variance of the data (onto the subset of dimensions).

PCA is one way for emphasizing variance and for revealing patterns of significance in a set of data. It does this by transforming data into new components, a collection of linearly uncorrelated variables. The number of principal components is less than the number of the original variables or independent variables.

PCA simplifies the dataset while retaining as much information as possible

Both factor analysis and principal component analysis are dimension reduction techniques. Factor analysis has factors, and principal component analysis has principal components. Principal components from PCA are generated from a linear combination of the variables of the original dataset. The purpose of PCA is to find no of principal components that explains as much of the cumulative variance of the original variables. Factor analysis is a model to measure factors or latent variables. The purpose of factor analysis is to find latent variables that can best explain the covariance among original variables, not the total variance of the dataset.

PCA involves the following steps mathematically:

1. **Centering the Data:** Subtract the mean of each feature to center the dataset:

$$X_{\text{centered}} = X - \mu$$

where X is the original dataset, and μ is the mean vector of the features.

2. **Covariance Matrix:** Compute the covariance matrix:

$$\Sigma = 1/n - 1 (X_{\text{centered}})^T X_{\text{centered}}$$

where Σ is the covariance matrix of size $p \times p$ (where p is the number of features).

3. **Eigen Decomposition:** Compute the eigenvalues (λ) and eigenvectors (v) of the covariance matrix:

$$\Sigma v_i = \lambda_i v_i$$

4. **Projection:** Project the original data onto the top k principal components

$$Z = X_{\text{centered}} V_k$$

where V_k is a matrix containing the top k eigenvectors as columns, and Z is the transformed dataset in the reduced space.

Category 1:

Lasso Regression

Lasso (Least Absolute Shrinkage and Selection Operator) regression is a linear regression method that includes L1 regularization.

Lasso regression can be used for feature extraction. This is a supervised method

- If $\alpha = 0$, then a lasso regression estimation is an OLS estimation.
- Lasso regression is a good choice when we have too many independent variables and we are looking to select only key variables for the model.

Lasso regression can first group independent variables with high correlations together, choose only one or few of them for the model, and drop the rest.,

$$L = 1/2n \sum_{i=1}^n (y_i - X_i^T \beta)^2 + \alpha \sum_{j=1}^p |\beta_j|$$

where $1/2n \sum_{i=1}^n (y_i - X_i^T \beta)^2$ is the mean square error

$\alpha \sum_{j=1}^p |\beta_j|$ is added penalty proportional to the absolute values of the coefficients. The penalty term can shrink some coefficients to zero thus performing feature extraction

Step 2

Category 3:

Principal Component Analysis (PCA):

PCA works by

1. Standardizing each dimension in the dataset, so all features have mean of 0 and standard deviation of 1,
2. Calculating the covariance matrix, and breaking down the matrix into eigen-values and eigenvectors. Sort the eigen-values in the decreasing order and choose the top k eigen-values and corresponding eigen vectors. This would help to form new dimensions, with the number of principal components (k) to be based on the percentage of variance to be retained in the data (like 95%).
3. The original dataset is then transformed onto the new feature space defined by the chosen principal components; The new principal components are a linear combination of original features.

Standardized Variable = $(\text{Original variable} - \text{Mean}) / \text{Standard deviation}$

Covariance = $\sum (x_i - \text{mean}(x)) * (y_i - \text{mean}(y)) / N$

From our two-variable example, a covariance matrix for two variables will be a 2 X 2 covariance matrix as below:

$\begin{pmatrix} \text{Cov}(\text{Var1}, \text{Var1}) & \text{Cov}(\text{Var2}, \text{Var1}) \\ \text{Cov}(\text{Var1}, \text{Var2}) & \text{Cov}(\text{Var2}, \text{Var2}) \end{pmatrix}$

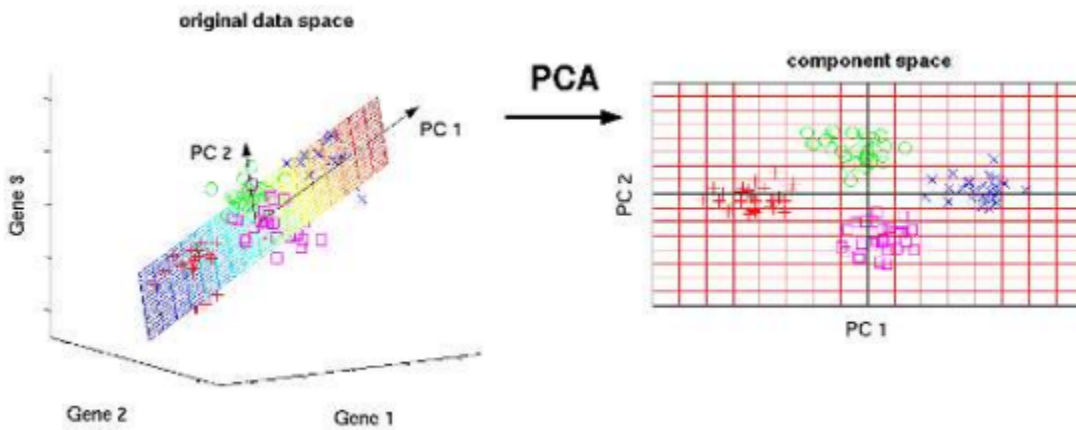
where the diagonal elements of matrix $\text{Cov}(\text{Var1}, \text{Var1})$ and $\text{Cov}(\text{Var2}, \text{Var2})$ are the variance.

While doing PCA, we can choose to have fewer principal components, without losing a lot of information, so this is also known as dimension reduction technique. Too many dimensions can cause issues like cramming and not generalizing on the data, overfitting and performance issues.

The big advantage of PCA is that the individual principal components are independent from each other. i.e. the principal components in the new coordinate system all point in exactly orthogonal directions to each other geometrically, so they are statistically uncorrelated. If we choose to use principal components as our new coordinate system, then we will not have the issue of multicollinearity in a regression model.

$PC1 = a_1 \text{Var1} + a_2 \text{Var2}$

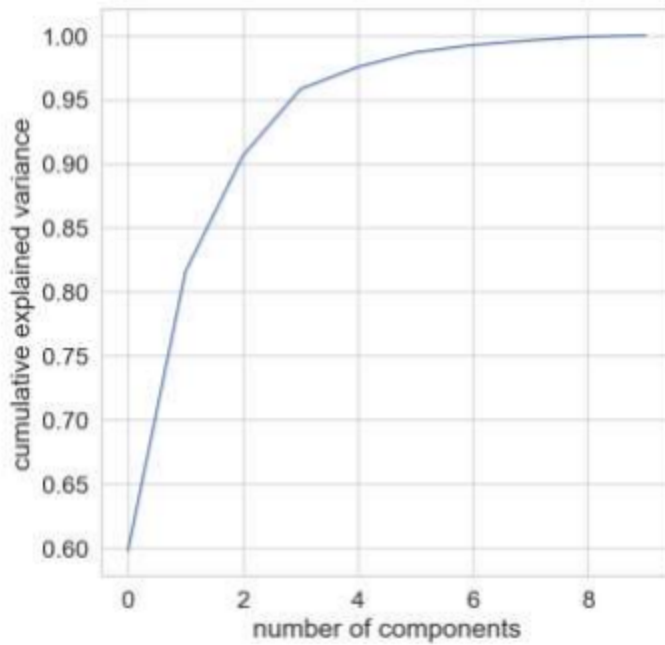
$$PC2=b1Var1+b2Var2$$



Hyperparameter tuning:

- Number of Components (k): Higher k means more variance is retained but less dimensionality reduction.
- Use the cumulative explained variance ratio to choose k such that a high proportion of variance (e.g., 95%) is retained.
- Perform grid search to evaluate the impact of different numbers of components on a downstream model's performance.

For e.g. in our case we imported data for ten stocks from 2019 to 2024. We applied PCA to the original dataset and it gave four principal components that can explain 95% of cumulative variance

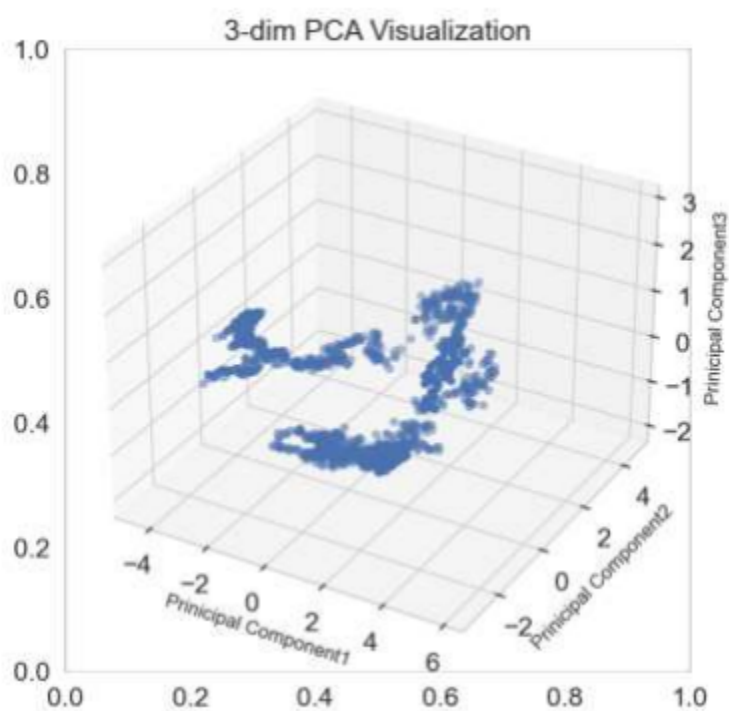
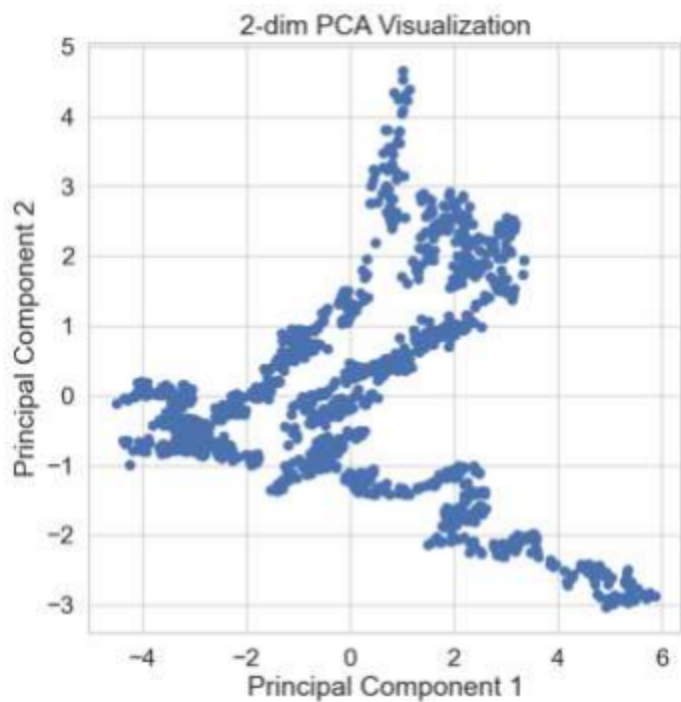


For each four rows, in the below PCA table of coefficients, we select the observation with highest no of absolute coefficients as proxy for that principal component.

	AAPL_Adj_Close	MSFT_Adj_Close	META_Adj_Close	FORD_Adj_Close	GM_Adj_Close	TSLA_Adj_Close	AMZN_Adj_Close	GOOG_Adj_Close	IBM_Adj_Close
0	0.387852	0.392648	0.327119	0.043803	0.213634	0.330359	0.317017	0.395751	0.302085
1	-0.072053	-0.142170	-0.002953	0.634237	0.463476	0.212475	0.274909	0.023804	-0.378039
2	-0.413816	-0.147139	0.558788	0.084132	0.216951	-0.499685	0.193193	-0.151898	0.108119
3	0.045186	0.079482	0.334467	-0.152952	-0.563912	-0.000717	0.580906	-0.050941	-0.319625

Thus, we got 'GOOG_Adj_Close', 'FORD_Adj_Close', 'META_Adj_Close', 'AMZN_Adj_Close' as our four important feature

2-dim and 3-dim PCA visualization as scatter plots.



PCA can have several issues:

1. Non Standardized data can lead to misleading principal components, we need to make sure data is standardized otherwise, it will get affected by large numbers or scales or different units,
2. We must check there is enough cumulative variance explained by new features and we select the right number of principal components. Choosing too few components can result in loss of information and too many can make the model complex and not help us in dimensional reduction.
3. We should check multicollinearity, address outliers which can be due to certain anomalies and not the real pattern of data. Multicollinearity can lead to redundant information and overemphasize certain patterns.
4. PCA is unsuitable for non-linear relationships or patterns, and using pca in this case may not give good model performance.
5. If PCA is not helping in improving the model performance, then we should consider changing the model or model hyper parameters or other techniques like factor analysis

Factor Analysis:

Factor Analysis can help in dimensional reduction and feature extraction. It is based on assumption that there are underlying unobserved latent variables called factors and these latent variables are linearly related observed variables called indicators.

$$Y_1 = B_{11} * F_1 + B_{12} * F_2 + \dots + B_{1m} * F_m + e_1 \dots \dots \dots$$

$$Y_n = B_{n1} * F_1 + B_{n2} * F_2 + \dots + B_{nm} * F_m + e_n.$$

There are n indicators and m factors. The goal of factor analysis is to determine the optimal number of factors to explain the variables. There are two types of Factor analysis: Exploratory Factor Analysis (EFA) and Confirmatory Factor Analysis (CFA)

There are two types of rotation methods in Factor Analysis. 1. Varimax rotates the axes to make the factors uncorrelated. 2. Promax rotation allows factors to be correlated.

Category 1:

Lasso Regression

Lasso regression has the following penalty function: $f() = \sum_{j=1_to_p} |\beta_j| = \|\beta\|$

The penalty function is called an L1 penalty function.

The function is based on the absolute value of the magnitude of coefficients in the model. Lasso regression can shrink the coefficients of the independent variables that don't contribute very much to the model to 0. Hence, lasso regression can help in feature extraction.

Hyper parameter tuning:

alpha: Regularization strength.

- alpha=0: Reduces to ordinary least squares regression.
- Larger alpha: Increases regularization, shrinking more coefficients toward zero.

1. Grid Search

Perform an exhaustive search over a predefined range of alpha values to find the one that minimizes the error metric.

2. Randomized Search

Similar to Grid Search but samples hyperparameters randomly. Useful for large parameter spaces.

3. Cross-Validation with LassoCV

Use LassoCV, which performs cross-validation internally to find the optimal alpha.

We got alpha as `lasso_cv.alpha_`: 31.622

	OLS	Ridge	Lasso
AAPL_Adj_Close	-0.025287	-0.025286	-0.000000
MSFT_Adj_Close	0.119807	0.119807	-0.012945
META_Adj_Close	0.335860	0.335860	0.334539
FORD_Adj_Close	10.208091	10.208082	0.000000
GM_Adj_Close	-0.222397	-0.222397	0.000000
TSLA_Adj_Close	0.110566	0.110566	0.135114
GOOG_Adj_Close	-0.185415	-0.185415	-0.000000
IBM_Adj_Close	0.026859	0.026859	-0.000000
GE_Adj_Close	-0.585318	-0.585318	-0.485793
R sq	0.916053	0.916053	0.887084
Mean Sq. Err	9.293714	9.293714	10.778629

Lasso Penalized Regression has made the coeff of Apple, Ford, Google, IBM, GM to equal to zero,, reducing the nine variables to four...thus helping in feature extraction..and coeff. of MSFT is also close to zero i.e. -0.013 and Lasso is giving R-squared as 89%..whereas Ridge has given R squared as 91%..which is good.

Step 3

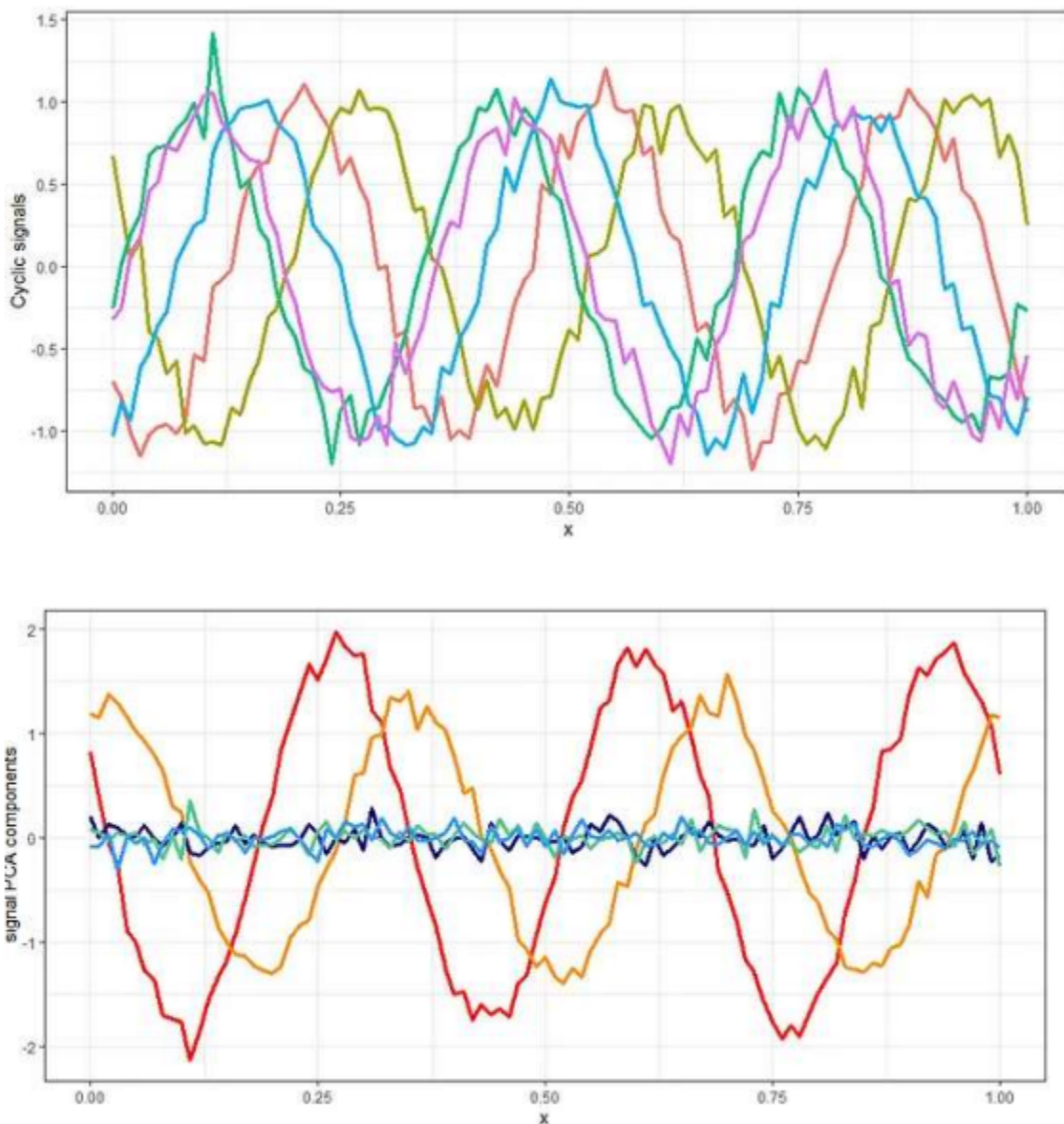
Category 3: PCA

Principal Component Analysis (PCA) is a method in machine learning used to lower down the dimensions of the datasets, by focusing on the most important aspect or variance.

PCA is an unsupervised dimension reduction technique, used with datasets having a large number of dimensions. It brings down the size of dimensions without considering for class labels, resulting in feature extraction. It reduces complexity, noise reduction and results in improved performance. This can be really useful when we're trying to figure out what's going on in our data and want to make things simpler. PCA takes correlated variables (tend to move together) and turns them into linearly uncorrelated principal components (PCs). These new PCs explain most of the variance from original data. It's esp. useful when we are working with high dimensional data like images or genetic sequences.

For e.g below is the image of five noisy signals,

After applying PCA, results in the reduced no of new variables like below



Advantages of PCA

1. **Dimensionality Reduction:**
 - o Reduces the number of features while retaining as much variance as possible, simplifying data analysis and visualization.
2. **Noise Reduction:**
 - o By capturing the most significant patterns in the data, PCA can filter out noise (low-variance components).
3. **Improves Algorithm Performance:**
 - o Many machine learning algorithms perform better with fewer features due to reduced computational cost and the curse of dimensionality.
4. **Decorrelates Features:**
 - o Transforms correlated features into a set of uncorrelated principal components, making the data more suitable for algorithms that assume feature independence (e.g., linear regression).
5. **Visualization:**
 - o Enables visualization of high-dimensional datasets in 2D or 3D by projecting them onto the top 2 or 3 principal components.
6. **Feature Extraction:**
 - o Identifies underlying patterns in the data by combining original features into principal components.

Disadvantages of PCA

1. **Loss of Interpretability:**
 - o Principal components are linear combinations of the original features, which can make them hard to interpret.
2. **Assumes Linearity:**
 - o PCA captures linear relationships between features. It may not perform well if the data has complex, non-linear patterns.
3. **Sensitivity to Scaling:**
 - o PCA is highly sensitive to the scale of the features. Features need to be standardized or normalized before applying PCA.
4. **Focus on Variance:**
 - o PCA maximizes variance, which may not always align with preserving the most relevant features for a specific task.
5. **All Components are Retained Initially:**
 - o The process requires computation of all eigenvectors and eigenvalues, which can be computationally expensive for large datasets.
6. **Potential Overfitting:**
 - o If too many components are retained, PCA may not significantly reduce dimensionality, leading to potential overfitting.
7. **Not Robust to Outliers:**
 - o Outliers can disproportionately affect the covariance matrix and the principal components.

Factor Analysis:

- Standardize the data
- Check for correlations in the times series
- Conduct Bartlett's Test and Kaiser-Meyer-Olkin (KMO) Test
- Create 8-factor object and fit data
- Visualize eigenvalues on scree plot
- Results of eigenvalues and visualized Scree plot help us decide number of factors. In the Scree plot, if we see that 3 eigenvalues are greater than 1 which means that we need to choose a 3-factor model.
- Create 3-factor object and fit data
- Get Factor Loadings
- Get proportion variance and cumulative proportion variance
- Create 3-factor object with varimax or promax rotation and fit data

Category 1:

Lasso Regression

Advantages of Lasso Regression

1. **Feature Selection:**
 - Lasso performs automatic feature selection by shrinking some coefficients to exactly zero, effectively removing irrelevant features.
2. **Simple and Interpretable Models:**
 - By eliminating non-informative features, Lasso creates simpler models that are easier to interpret.
3. **Prevents Overfitting:**
 - The L1 regularization penalizes large coefficients, helping reduce overfitting and improving generalization to unseen data.
4. **Handles Multicollinearity:**
 - Lasso can handle multicollinear data by selecting one feature from a group of highly correlated features and shrinking others to zero.
5. **Efficient:**

- o Computationally efficient for datasets with many features, as it reduces dimensionality during model training.
- 6. **Good for Sparse Models:**
 - o Works well when the true underlying model is sparse (i.e., only a few features are truly relevant).

Disadvantages of Lasso Regression

1. **Biased Estimates:**
 - o The L1 regularization can introduce bias in the estimated coefficients, especially for larger values of the regularization parameter α .
2. **Feature Selection Instability:**
 - o The features selected by Lasso can vary significantly with small changes in the data or the regularization parameter.
3. **Limited with High-Dimensional Data:**
 - o In cases where the number of features p is much greater than the number of observations n , Lasso may select too many variables or fail to select the correct ones.
4. **Single Feature Selection from Correlated Groups:**
 - o When features are highly correlated, Lasso tends to select only one feature from the group, which might not always be the most meaningful or representative.
5. **Requires Tuning:**
 - o The hyperparameter α (regularization strength) requires careful tuning to balance bias and variance.
6. **Not Suitable for All Problems:**
 - o Lasso might not perform well if all features contribute equally to the target variable, as it is designed to create sparse models.
7. **Difficulty with Non-Linear Relationships:**
 - o Lasso assumes a linear relationship between predictors and the target variable, making it unsuitable for capturing complex, non-linear patterns without additional transformations.

Lasso Regression Hyperparameter Tuning:

Primary Hyperparameter: α (alpha) - Regularization Strength

$\alpha = 0$: Reduces to ordinary least squares regression

Larger α : Increases regularization, shrinking more coefficients toward zero

Methods of Tuning:

a) Grid Search:

- Conduct an exhaustive search over a predefined range of alpha values
- Objective: Minimize error metric
- Example: In our analysis, we found $\alpha = 31.622$ using LassoCV

b) Randomized Search:

- Samples hyperparameters randomly
- Large parameter spaces
- Computational efficiency against grid search

c) Cross-Validation (LassoCV):

Does internal cross-validation to find the optimum alpha; helps in choosing the best regularization strength.

Step 4

The challenges that are associated with PCA (Principal Component Analysis) in data analysis, can be resolved by these clear, specific, and practical steps:

1. Standardize the Data (Challenge: Non-standardized data can lead to misleading principal components, as the results are affected by large numbers, scales, or different units)

Directions to resolve it:

- ❖ Standardize the Data:
- ❖ Use statistical software (e.g., R, Python, Excel) to standardize your data before applying PCA.
- ❖ Standardization involves rescaling the data to have a mean of 0 and a standard deviation of 1.

2. Select the Appropriate Number of Principal Components (Challenge: Choosing too few components can result in a loss of information, while choosing too many can make the model complex and not aid in dimensional reduction)

• Directions to resolve it:

- ❖ Determine Cumulative Variance:
- ❖ Calculate the cumulative variance explained by each principal component.
- ❖ Select the number of components that explain a sufficient amount of variance (e.g., 95%).

3. Check for Multicollinearity and Address Outliers (Challenge: Multicollinearity can lead to redundant information and overemphasize certain patterns. Outliers can distort the principal

components and lead to incorrect conclusions.)

Directions to resolve it:

- ❖ Check for Multicollinearity:
- ❖ Calculate the correlation matrix of your original features to identify highly correlated variables.
- ❖ Remove or combine highly correlated variables before applying PCA.
- ❖ Detect and Treat Outliers:
- ❖ Use statistical methods (e.g., Z-scores, IQR) and visual tools (e.g., box plots) to detect outliers.
- ❖ Remove or transform outliers to reduce their impact.

4. Assess Suitability for Non-Linear Relationships (Challenge: PCA is unsuitable for non-linear relationships or patterns, leading to poor model performance.)

Directions to resolve it:

- ❖ Use Kernel PCA for Non-Linear Data:
- ❖ If your data exhibits non-linear patterns, consider using Kernel PCA, which can capture non-linear relationships.
- ❖ Evaluate Model Performance:
- ❖ Compare the performance of models using linear PCA and Kernel PCA to determine which approach better captures the underlying data structure.

5. Consider Alternative Techniques

Directions:

- ❖ Explore Other Dimensionality Reduction Techniques:
- ❖ If PCA does not yield satisfactory results, consider using Factor Analysis, t-SNE, or UMAP.
- ❖ Adjust Model Hyperparameters:
- ❖ Tune the hyperparameters of your machine learning model to improve performance with the reduced data.
- ❖ Use techniques like grid search or random search for hyperparameter optimization.

The issues caused by PCA can be addressed with the above-mentioned directions

Sometime even after we applied the dimension reduction technique like vif or pca, the model has less R-squared, reason could be that the model failed to generalize on the data like there could be very weak relationship or no relationship at all. In those cases we may need to look for other techniques like ensemble methods or use some domain specific algorithms or models suited for the use case.

The challenges that are associated with the **Lasso Regression** can be solved using the practical steps as following:

Manage Multicollinearity Problem: High correlation among predictors might lead to the instability of estimates of coefficients, hence affecting the model interpretability of the model.

Directions for Solution :

❖ Analysis of Correlation:

- Calculate the correlation matrix to spot highly correlated predictors.
- Handpick or combine the features that are highly correlated while applying Lasso Regression.

❖ Feature Selection Strategy

- Employ domain knowledge in selecting meaningful predictors.
- Consider removing or clubbing redundant features to help in improving model stability.

Handle Complexity and Regularization of the Model Challenge: The challenge may involve choosing an appropriate regularization parameter, which if not correctly chosen, could lead to either underfitting or overfitting.

Solution

There are a number of ways through which one can handle the model complexity and regularization, including:

Cross-Validation Techniques:

- Implement k-fold cross-validation in determining the best lambda (regularization parameter).
- Grid search and coordinate descent methods for best lambda value

Performance Evaluation

- Comparison of performance evaluation using Mean Squared Error-MSE or Mean Absolute Error-MAE.

- Plot the Regularisation Path: It helps in knowing how coefficients vary under various values of λ .

Handling Sparse and Small Datasets: The challenge is that Lasso may not work well in cases of small datasets, when the number of predictors either equals or is less than the number of observations.

Steps for Solution:

- Data Augmentation
- Collect more data, wherever possible, to increase the sample size.
- Do Bootstrapping to generate Synthetic Samples.

Alternative Regularization:

- Ridge Regression or Elastic Net handles small datasets better.
- Implement careful feature selection based on domain expertise.

By systematically addressing these challenges, Lasso Regression can be effectively implemented and its strengths in feature selection and regularization leveraged.

Step 5

PCA (Principal Component Analysis) is one way to extract features from data. It can be sensitive to different units or large feature scales. We should check the mean (as 0) and standard deviation (as 1) of all the features. We need to make sure our data is standardized, and use tools like summary statistics or box-plots. We can use the sklearn library and `StandardScaler` function to help standardize our data.

To make sure PCA is capturing important information, we should calculate the cumulative explained variance (like 95%) and use a scree plot to test it. This helps us find the optimum number of principal components.

PCA can handle multicollinearity in features by transforming features of original data into uncorrelated new features called components. There can be an issue, if there is high multicollinearity, we might need to do feature engineering. To detect multicollinearity, we should compute a correlation matrix and use seaborn heatmap visualization.

The number of principal components we choose for feature extraction is important, because it affects how many features and how well they explain the data. We want to choose a number that explains a lot of the cumulative variance in our data. We may not choose too few components, that may result in loss of information and not capturing enough variance and including too many components don't help us in the purpose of dimensional reduction.

Outliers can affect our results, so we have to identify and assess them using scatter plots or boxplots. If our data has non-linear relationships, we should consider using alternative dimensionality reduction techniques. We can use residual plots to check how well these methods are working compared to PCA.

PCA is primarily designed for continuous data. For categorical variables, alternatives like Multiple Correspondence Analysis (MCA) or embeddings are more appropriate.

PCA vs. Alternatives Methods

1. **PCA:**
 - o For linear datasets, noise reduction, or when variance is the primary concern.
2. **Kernel PCA:**
 - o For non-linear datasets where relationships are not captured by standard PCA.
3. **t-SNE/UMAP:**
 - o For visualizing clusters or manifold structures in high-dimensional data.
4. **Autoencoders:**
 - o When dealing with large, non-linear datasets that need flexible representations.
5. **LDA:**
 - o When dimensionality reduction and classification are both required.

Aspect	PCA	t-SNE	Kernel PCA	Autoencoders	LDA
Linear vs. Non-linear	Linear	Non-linear	Non-linear	Non-linear	Linear
Supervised	No	No	No	Both	Yes
Feature Interpretability	No	No	No	No	Yes
Scalability	High	Low	Medium	Medium	Medium
Outlier Sensitivity	High	Moderate	High	Moderate	High

Category 1:

Lasso Regression

Alternative methods than Lasso:

Ridge

Regularization L2-regularization (β^2)

Feature Selection Shrinks coefficients but doesn't make them exactly zero.

Multicollinearity Handles multicollinearity well by distributing the effect among correlated features.

Use Case When all features are important.

Interpretability Retains all features. .

Elastic Net combines the strengths of Ridge and Lasso, allowing both regularization types. It's highly useful when working with datasets that have correlated features and require feature selection.

Disadvantages of Elastic Net

1. **Increased Complexity:**
 - o Requires tuning two hyperparameters: α (regularization strength) and ρ (mixing parameter).
2. **Bias in Coefficients:**
 - o Like Ridge and Lasso, Elastic Net introduces bias in the coefficient estimates due to regularization.
3. **Not Optimal for Sparse or Dense Data Alone:**
 - o For purely sparse data, Lasso might be better. For purely dense data, Ridge could outperform Elastic Net.

References

1. Author: Anshul, Dimensionality Reduction using PCA on multivariate timeseries data, Dated: Sep, 2019
[Dimensionality Reduction using PCA on multivariate timeseries data | by Anshul | Medium](#)
2. Author: John Mark Agosta, The PCA Trick with Time-Series, Dated: Dec, 2020
[The PCA Trick with Time-Series. PCA can be used to reject cyclic... | by John Mark Agosta | Towards Data Science](#)

3. .Author:Pletskin, Multivariate Seasonal Timeseries Prediction using Principal Component Analysis, Dated: 2019
[GitHub - ptleskin/MultivariateSeasonalTimeseriesPredictionPCA: Multivariate Seasonal Timeseries Prediction with PCA](#)